

Murtaza Bootwala

Aspiring Computer Engineer and Robotics Enthusiast

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Aspiring computer engineer with hands-on experience in robotics fabrication and effective instruction, blending technical proficiency with a passion for team development and problem-solving within dynamic environments.

PROFESSIONAL EXPERIENCE

Aero and Chassis Team | Cal Poly Racing

Sep 2025 – Present

San Luis Obispo, CA

- Assisted Aero & Chassis subgroups in manufacturing components for the Formula SAE vehicle, contributing to a 10–15% reduction in composite fabrication time through improved material handling
- Performed carbon layups, vacuum bagging, material preparation, and post-processing with <2% rework rate, ensuring quality and consistency in composite parts
- Assisted in prepping and assembling the carbon chassis tub with a focus on quality, accuracy, and structural performance
- Gained experience in composite fabrication workflows, safety procedures, and collaboration in a multidisciplinary engineering environment

Fabrication Lead | FRC Team 2976, Spartabots

Aug 2023 – Jun 2024

Sammamish, WA

- Guided a team of 10 members through detailed robotics fabrication tasks, ensuring all parts met strict specifications under competition deadlines
- Designed and generated CAD models, translating design concepts into machine-operable instructions
- Operated CNC machinery and oversaw the transformation of raw materials into precise components
- Managed workflow and quality assurance processes while promoting strong communication throughout the team
- Troubleshoot fabrication challenges in real time, minimizing delays and maintaining optimal productivity

Swim Instructor | YMCA of Seattle

Jun 2023 – Sep 2025

Sammamish, WA

- Delivered tailored swim lessons to students across age groups and skill levels, adapting instructional techniques for effective knowledge transfer
- Monitored class safety, ensured adherence to established safety guidelines, and intervened promptly in emergency situations
- Cultivated an engaging and supportive environment that encourages student growth and builds water confidence
- Assessed individual progress and provided constructive feedback to help learners refine technique
- Coordinated scheduling and lesson plans with colleagues to maintain consistently high instructional standards

SKILLS

- Composite Fabrication (Carbon Fiber, Vacuum Bagging)
- Mechanical Fabrication & Assembly
- CNC Machine Operation
- CAD Modeling & Technical Drawings
- Design for Manufacturing
- Robotics Fabrication & Prototyping
- Manufacturing Process Optimization
- Manufacturing Safety & Compliance
- Java (JavaFX)
- Python
- HTML/CSS & JavaScript
- Cloud Computing Fundamentals (AWS)
- Troubleshooting & Problem Solving
- Team Leadership & Mentorship
- Technical Instruction & Communication
- Hands-On Prototype Testing & Validation
- Failure Analysis

EDUCATION

California Polytechnic State University

San Luis Obispo, CA

Bachelor's degree, Computer Engineering • Expected Graduation: June 2029

CERTIFICATES

- AWS Certified Cloud Practitioner | Amazon Web Services Training and Certification — August 15, 2025
- Harvard AI Bootcamp | Harvard University — December 2023
- Networking Basics | Cisco — March 27, 2025